

Skills Summary

Ratios, Inverses and Calculator Mode

Use this reference while completing your worksheet or when studying.

Skill 1 — Choosing the right ratio

Name the two sides first, then the ratio follows. With the angle A fixed, label the sides **opp**, **adj** and **hyp**:

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}.$$

Tangent never involves the hypotenuse, so it can neither find one nor use one.

Example. A right triangle has hypotenuse 18 and one acute angle 41° . Find the side adjacent to that angle.

Step 1. Want the adjacent side, know the hypotenuse — adjacent over hypotenuse is cosine, and the unknown is on top so I multiply.

Step 2. $\text{adj} = 18 \cos 41^\circ = 13.5847 \dots$

$\text{adj} \approx \mathbf{13.58}$

Try it. A right triangle has hypotenuse 25 and one acute angle 63° . Find the side opposite that angle.

check: 22.28

Skill 2 — Choosing the right inverse

Rule. Two sides are known and the angle is not: form their ratio, then undo it with the matching inverse.

- opp and adj $\rightarrow \tan^{-1}$
- opp and hyp $\rightarrow \sin^{-1}$
- adj and hyp $\rightarrow \cos^{-1}$

Dividing alone gives the *ratio*, not the angle. The inverse function is what converts one into the other.

Example. A right triangle has a side of 9 opposite $\angle A$ and a side of 14 adjacent to it. Find $m\angle A$.

Step 1. The two known sides are opposite and adjacent, which is the tangent pairing — so the angle is the inverse tangent of their quotient.

Step 2. $A = \tan^{-1}(9/14) = 32.7352 \dots^\circ$

$m\angle A \approx \mathbf{32.74}$ degrees

Try it. A right triangle has a side of 5 adjacent to $\angle A$ and a hypotenuse of 12. Find $m\angle A$.

check: 59.88 degrees

Skill 3 — Degree mode and a size check

Two habits that catch a mode error every time.

- Confirm the calculator shows DEG before the first line of work. In radian mode an angle such as 52 is read as 52 radians, and the result can even come out negative — impossible for a side length.
- Estimate before you accept. A leg is always shorter than the hypotenuse; a 45° angle splits a right triangle evenly, so a smaller angle means a shorter opposite side and a longer adjacent one.

Example. A right triangle has a side of 30 adjacent to a 25° angle. Find the opposite side, and check the size of the answer.

Step 1. Opposite over adjacent is tangent, and the unknown is on top — multiply, with the calculator in degree mode.

Step 2. $\text{opp} = 30 \tan 25^\circ = 13.9892\dots$, and since 25° is well under 45° the opposite side should indeed be well under 30.

$$\text{opp} \approx \mathbf{13.99}$$

Try it. A right triangle has a side of 16 adjacent to a 40° angle. Find the opposite side.

check: 13.43